

Technology #03

Catalytic Contact Time Lock for Amyloid Beta
Clearance

Engineering Technology-Transfer Dossier
CONFIDENTIAL

Technology Maturity: ENGINEERING_DEFINITION
Generated: 2026-08-27

Not physically validated. Not transfer-ready.

BUYER DECISION PAGE

WHAT IS IT?

Catalytic Contact Time Lock for Amyloid Beta Clearance

WHY DOES IT MATTER?

Amyloid beta accumulation in CSF of Alzheimer's/NPH patients with shunts

WHAT IS ACTUALLY ESTABLISHED?

Engineering definition with 5 governing equations, 11 design inputs, 5 failure modes. Computational model exists. No physical prototype.

WHAT IS NOT ESTABLISHED?

Physical validation, manufacturing process, regulatory pathway, IP ownership.

WHY MIGHT A BUYER CARE?

Opportunity in Enzymatic Catalysis + Mass Transport. 6-step build plan with specific test articles and acceptance criteria. Target buyer: Biotech company with enzyme therapy program; neuroshunt manufacturer.

WHAT DOES THE BUYER GET?

Conceptual design + governing equation (Michaelis-Menten + Damköhler); Mechanism description (immobilized NEP on catheter lumen); Bench test protocol (clearance per pass, stability); External precedent catalog (NEP kinetics, immobilization chemistry); This engineering dossier with failure analysis + build plan

WHAT DOES THE BUYER HAVE TO BUILD?

Recombinant NEP production process (GMP-grade); Coating process specification + QC; Final catheter geometry (length, diameter, coating density); Manufacturing process for enzyme-active catheter (aseptic assembly); Regulatory submission (likely Class III PMA — no predicate for enzymatic catheter); Clinical validation (CSF A β 42 reduction + clinical benefit endpoint); Long-term stability and reliability data

WHAT IS THE NEXT DECISIVE EXPERIMENT?

Recombinant NEP enzyme lot

WHAT WOULD MAKE US KILL IT?

Enzyme half-life in CSF environment is less than 30 days (insufficient for chronic implant)

WHAT TRANSACTION COULD MAKE SENSE?

Sponsored Validation

What Exists Today

Existing:

- + Documented mechanism with governing equations
- + Computational model
- + 5 external evidence sources
- + Design inputs and outputs
- + Failure analysis
- + 6-step engineering build plan
- + Transfer boundary definition

Does Not Yet Exist:

- Physical prototype
- Qualified manufacturing process
- Clinical data
- Verified IP ownership
- Regulatory clearance
- Commercial product

Buyer Can Commission:

- > Specific bench experiment (see build plan)
- > Prototype fabrication
- > Materials study
- > Manufacturing feasibility study
- > Regulatory pre-submission

Development / De-risking Ladder

CURRENT STATE: ENGINEERING_DEFINITION

NEXT ENGINEERING STEP: Recombinant NEP enzyme lot

NEXT VALIDATION STEP: Bench prototype validation per build plan

NEXT REGULATORY STEP: Pre-submission to FDA for pathway classification

COMMERCIALIZATION GATE: Clinical validation evidence required

TRANSFER GATE: Transfer-ready when physical validation + regulatory pathway + IP verified

Engineering Investment Ladder

Next \$5K action: WP-01 - Recombinant NEP enzyme lot - 12 weeks

Basis: Build plan WP. Scope: k_{cat} , Km, purity, sterility. Uncertainty: Effort estimate, not quoted.

Next \$25K action: WP-03 - NEP-coated catheter segment - 6 weeks

Basis: Build plan WP. Scope: Clearance per pass at various Q. Uncertainty: Effort estimate, not quoted.

Next \$100K action: WP-05 - NEP-coated specimens - 12 weeks (external lab)

Basis: Build plan WP. Scope: Cytotoxicity, sensitization, irritation, systemic toxicity. Uncertainty: Effort estimate, not quoted.

Top 5 Buyer Diligence Questions

Q1: Is the mechanism physically credible for the intended application?

The governing model includes 5 equations. Key assumptions: NEP retains activity when immobilized (external precedent, NOT verified for this coating); A β 42 is the rate-limiting substrate (other peptides may compete; UNKNOWN); CSF A β 42 concentration in catheter inlet matches published ranges; Enzyme surface density Gamma is uniform (process-dependent); No significant product inhibition (UNKNOWN for A β fragments). UNKNOWN: whether assumptions hold in vivo.

Q2: What evidence supports the core claim?

5 external evidence sources. All are external precedent, not physical validation of this specific invention.

Q3: What is the regulatory pathway?

UNKNOWN. Not yet determined. Requires pre-submission to FDA. No predicate identified for some packages.

Q4: Who owns the IP?

UNKNOWN. IP ownership not verified. Requires legal counsel engagement.

Q5: What would it cost to validate?

First experiment: Recombinant NEP enzyme lot. Estimated effort from build plan. Full validation cost: COST TO BE QUOTED.

Commercial Interest

TARGET APPLICATION: CSF shunt systems for hydrocephalus management

CURRENT ALTERNATIVE: Standard fixed-pressure or programmable CSF shunt valves

MECHANISM DIFFERENCE: NEP enzyme immobilized on catheter wall cleaves A-beta-42 via mass-transport-limited contact time

POTENTIAL VALUE DRIVER: Addresses: Amyloid beta accumulation in CSF of Alzheimer's/NPH patients with shunts

ADOPTION BARRIER: Requires physical validation, regulatory clearance, and manufacturing qualification

DEVELOPMENT BARRIER: No physical prototype exists; manufacturing process not established

Licensee Capability Fit

IDEAL BUYER: Biotech company with enzyme therapy program; neuroshunt manufacturer

REQUIRED ENGINEERING CAPABILITY: Catheter design, fluid mechanics / domain-specific engineering

REQUIRED MANUFACTURING CAPABILITY: Medical tubing extrusion, injection molding, cleanroom assembly

REQUIRED REGULATORY CAPABILITY: FDA 510(k) or PMA submission experience

REQUIRED CAPITAL: Validation: \$50-200K. Full development: \$500K-2M (estimates, not quoted)

REQUIRED MARKET ACCESS: Neurosurgery market channels; hospital purchasing relationships

1. Technology Description

Domain: Enzymatic Catalysis + Mass Transport

Disciplines: Enzyme kinetics, Mass transport (diffusion + convection), Surface chemistry (enzyme immobilization), Biomaterials (catheter coatings)

Architecture: Catheter with immobilized neprilysin (NEP) enzyme on inner lumen wall; CSF flows past, A β 42 in CSF contacts enzyme for sufficient time (contact-time lock) to undergo catalytic cleavage

Subsystems:

SS-01: Catheter body (substrate) - Carries CSF flow + supports enzyme coating

SS-02: NEP-immobilized coating on inner lumen - Catalytic surface cleaves A β 42 peptides via NEP activity

SS-03: Contact-time control geometry - Catheter length and flow rate engineered so A β 42 residence time exceeds critical contact time for catalytic cleavage

SS-04: Stabilization matrix for enzyme - Preserves NEP activity over implantation lifetime (likely PEGylation or trehalose stabilization)

2. Mechanism Architecture

Physical Changes: A β 42 peptides in CSF contact immobilized NEP during catheter transit; NEP catalyzes proteolytic cleavage of A β 42 into inactive fragments, reducing CSF A β 42 burden

Key Physics: Mass-transport-limited enzyme catalysis: rate $v = \min(V_{\max} \cdot [S] / (K_m + [S]), J_{\text{conv}})$, where J_{conv} is convective delivery of A β 42 to the enzyme surface

3. Governing Engineering Model

Summary: Mass-transport-limited enzyme catalysis in catheter lumen. Reaction rate is bounded by either enzyme kinetics (Michaelis-Menten) or convective delivery of substrate to enzyme surface, whichever is smaller.

Equations:

$v = (V_{\max} * [S]) / (K_m + [S])$ [enzyme kinetics]

$J = -D * dC/dx$ [diffusive flux]

$Da = (k * L) / D$ [reaction vs transport]

$Pe = (v * L) / D$ [Péclet number, convective vs diffusive transport]

$k_{\text{cat_obs}} = k_{\text{cat}} * (1 - \text{deactivation}(t))$ [enzyme deactivation over time]

Assumptions:

- NEP retains activity when immobilized (external precedent, NOT verified for this coating)
- A β 42 is the rate-limiting substrate (other peptides may compete; UNKNOWN)
- CSF A β 42 concentration in catheter inlet matches published ranges
- Enzyme surface density Γ is uniform (process-dependent)
- No significant product inhibition (UNKNOWN for A β fragments)

Boundary Conditions:

- Inlet: A β 42 = 100-1000 pg/mL (pathological range, per literature)
- Outlet: A β 42 reduced by clearance fraction
- Wall: enzyme-bound reaction sink (Robin BC)
- Lumen: CSF flow at 0.3 mL/min typical

Failure Regimes:

- $Da \ll 1$ (transport-limited) — clearance is limited by delivery to enzyme; increasing Γ does NOT help

- $Da \gg 1$ (kinetics-limited) — clearance is limited by enzyme activity; increasing L does NOT help
- $\Gamma_t \ll \Gamma_{\text{initial}}$ (enzyme deactivation) — activity falls below useful threshold over time
- Competitive inhibition by other CSF peptides — UNKNOWN magnitude

4. Design Inputs

ID	Input	Value	Origin
DI-001	Clinical need	Aβ accumulation in CSF of Alzheimer's/NPH patients with shunts; no enzyme release exists (R332 problem)	VERIFIED
DI-002	Functional requirement	NEP enzyme immobilized on catheter wall clears Aβ42 via mass-transport-limited process	MODELLED
DI-003	CSF Aβ42 range (pathological)	100-1000 pg/mL (external literature, AD patients)	EXTERNAL_PRECEDENT
DI-004	Clearance target per pass	>= 30% (MODEL_DERIVED, R332)	MODELLED
DI-005	Enzyme kinetics (k_cat, Km)	EXTERNAL_PRECEDENT (published values, varies by assay)	EXTERNAL_PRECEDENT
DI-006	Flow rate (CSF through shunt)	~0.3 mL/min typical (R332)	VERIFIED
DI-007	Biocompatibility (ISO 10993)	UNKNOWN	UNKNOWN
DI-008	Sterilization	UNKNOWN — enzyme incompatible with gamma; likely EtO or aseptic assembly	UNKNOWN
DI-009	Enzyme source / manufacturing	UNKNOWN — recombinant NEP needed	UNKNOWN
DI-010	Coating uniformity / quality	UNKNOWN	UNKNOWN
DI-011	Shelf life (with active enzyme)	UNKNOWN — likely cold-chain dependent	UNKNOWN

5. Design Outputs

ID	Description	Status	Missing Inputs
DO-001	Catheter with NEP coating (length, ID, coating density)	ABSENT	Final length L; Coating process Gamma spec; Coating uniformity spec
DO-002	NEP coating process specification	ABSENT	Covalent vs adsorption chemistry; Curing protocol; QC release criteria
DO-003	Recombinant NEP production process	ABSENT	Expression system (E. coli vs CHO); Purification protocol; Lot release cr
DO-004	Clearance verification test method	CONCEPTUAL	

6. Critical Design Parameters

Name	Value	Unit	Basis	Verification
NEP k_cat	EXTERNAL_PRECEDENT (published values 10 ⁴ -10 ⁵ s ⁻¹ at 37°C by assay)	1/mol/s	Published NEP kinetics literature	Temperature independent assay on immobilized form
NEP Km for Aβ42	EXTERNAL_PRECEDENT (published ~5-20 μM range)	μM	Published NEP kinetics literature	Temperature independent assay on immobilized form
Enzyme surface density G_cat	UNKNOWN	mol/m^2	Design choice + coating process	Surface assay (e.g., fluorescent labeling, activity assay)
Catheter length L	UNKNOWN	m	Design choice balancing clearance and catheter size	Geometric specification
Lumen diameter d	UNKNOWN	m	Standard shunt catheter ~1-1.5 mm ID	Specification
Contact time tau	UNKNOWN	s	Derived: tau = L/v	Flow visualization
Enzyme half-life t_1/2	UNKNOWN	days	UNKNOWN for immobilized NEP	Animal study in CSF-mimicking fluid

7. Failure Modes

Mode	Mechanism	Mitigation	Verification	Residual Uncertainty
Enzyme deactivation	Protein denaturation, proteolysis, adsorption	Stabilization (PEGylation, lyophilization)	Stability studies in CSF mimicking fluid	Assuming stable half-life
Mass transport limit	If Da < 1, clearance limited by diffusion	Optimize L, d, v; use microfluidics	Calculate Da; in vitro diffusion tests	UNKNOWN actual Da in vivo
Immune response to enzyme	Host immune system recognition	PEGylation for immunosuppression	In vivo biocompatibility studies	UNKNOWN immunogenicity
Substrate competition	Other CSF peptides compete	Engineer high specificity	Competitive assays in CSF	UNKNOWN competitive magnitude
Coating delamination	Mechanical or chemical degradation	Covalent bonding; adhesion promoters	Adhesion tests; in vitro	UNKNOWN long-term adhesion

8. Failure Analysis

Failure Mode	Mechanism	Mitigation	Residual Uncertainty
Enzyme deactivation (chronic)	Protein denaturation, proteolysis, surface adsorption	Stabilization matrix + spare activity margin	UNKNOWN achievable half-life
Mass transport limit (Da < 100k)	Clearance limited by Aβ42 delivery, not by CSF turnover	Lengthen L or reduce Q (clinical trade-off)	UNKNOWN actual Da in vivo
Immune response to NEP	Host recognizes NEP as foreign	PEGylation or immunosuppressive coating	UNKNOWN NEP-specific immunogenicity
Substrate competition	Other CSF peptides compete for NEP	UNKNOWN — specificity engineering may be required	UNKNOWN competitive magnitude
Coating delamination	Mechanical/chemical bond failure	Covalent immobilization + mechanical testing	UNKNOWN long-term adhesion in vivo
Catheter obstruction (separate from CSF inlet)	Dextran sulfate-induced protein aggregation	Standard block solution mitigation + unknown change the yearly dextran coating	UNKNOWN catheter obstruction rate

9. Verification Strategy

ID	Requirement	Method	Acceptance	Result
V-001	NEP coating activity per unit area	Fluorescent substrate assay	Gamma * k_cat meets model requirements	NOT_TESTED
V-002	Clearance per pass >= 30%	Bench flow loop with Aβ42 mock CSF	>= 30% reduction	NOT_TESTED
V-003	Enzyme stability over implantation lifetime	Ageing study at 37°C in CSF mimic	Activity retained > 50% at target time	NOT_TESTED
V-004	Biocompatibility ISO 10993	Standard ISO 10993 series	Pass	NOT_TESTED

10. Validation Strategy

ID	Requirement	Method	Acceptance	Result
VAL-001	CSF Aβ42 reduction in patient cohort	Clinical trial (IDE required)	Statistically significant vs control	NOT_PERFORMED
VAL-002	Clinical benefit	Long-term trial	Unknown — no established endpoints	NOT_PERFORMED

11. Materials

Component	Candidate Material	Source	Verification Required	Status
Catheter body	Silicone elastomer	Standard CSF shunt material	ISO 10993 + coating adhesion	CANDIDATE
Catheter body	Polyurethane	Some shunts use polyurethane	ISO 10993 + coating adhesion	CANDIDATE
Enzyme coating	Recombinant human NEP (neprilysin)	Published NEP literature (PMC1251204, BioS 050229850)	Activity assay	CANDIDATE
Immobilization chemistry	Covalent EDC/NHS coupling	Standard protein immobilization	Coupling efficiency + leachability	CANDIDATE

12. Bill of Materials

Item	Description	Qty	Type	Material	Supplier	Criticality	Verification
01	Catheter body (substrate) 1	1	CUSTOM_COMPONENT	UNKNOWN — likely silicone or polyurethane	UNKNOWN	CRITICAL	Biocompatibility
02	Recombinant NEP enzyme	Micrograms	CONSUMABLE	Human NEP protein	UNKNOWN — recombinant	CRITICAL	Activity assay, purity + sterility
03	Coating chemistry reagent	Process	CONSUMABLE	EDC/NHS or similar coupling	Multiple (Sigma, Fisher)	HIGH	Coating QC
04	Stabilization matrix (e.g., PEG, trehalose)	Process	CONSUMABLE	PEG or trehalose	Multiple	HIGH	Stability study

13. Manufacturing

Status: ENGINEERING_CANDIDATE — no process validated for chronic enzyme-active catheter

Candidate Processes:

- Recombinant NEP production (E. coli or CHO expression): Standard recombinant protein manufacturing - Requires GMP manufacturer partnership
- Covalent enzyme immobilization on catheter inner lumen: Standard protein immobilization chemistry - Custom process development required
- Aseptic assembly (likely; enzyme incompatible with terminal sterilization): Standard aseptic processing - EtO may be evaluated but enzyme sensitivity UNKNOWN

14. External Evidence

Source 1: Neprilysin-Mediated Amyloid Beta Clearance and Its Therapeutic

URL: <https://pmc.ncbi.nlm.nih.gov/articles/PMC11651204>

Excerpt: Neprilysin (NEP) is a neutral endopeptidase, important for the degradation of amyloid beta (A β) cerebrospinal fluid (CSF) that is aimed at removing excess A β ...

Source 2: [PDF] Blood–brain barrier penetrating neprilysin degrades monomeric

URL: <https://uu.diva-portal.org/smash/get/diva2:1651329/FULLTEXT01.pdf>

Excerpt: Neprilysin (NEP) is a membrane-bound metallopeptidase and one of the major A β -degrading enzymes. position of amyloid plaques. Neprilysin-2 is an important β - ...

Source 3: Brain Shuttle Neprilysin reduces central Amyloid- β levels | PLOS One

URL: <https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0229850>

Excerpt: Mar 10, 2020 · As previously shown, NEP was highly effective at degrading A β in blood but not in the CSF compartment after systemic administration. In contrast ...

Source 4: Inhibiting Neprilysin Is Good for the Heart: What About the Brain?

URL: <https://www.alzforum.org/news/research-news/inhibiting-neprilysin-good-heart-what-about-brain>

Excerpt: Sep 12, 2014 · LCZ696 contains an inhibitor of neprilysin, an enzyme that breaks down the A β peptide that forms amyloid plaques in the brain.

Source 5: Advances in Amyloid- β Clearance in the Brain and Periphery

URL: <https://www.en-journal.org/journal/view.html?uid=633>

Excerpt: Aug 31, 2023 · The clearance of amyloid- β can occur through enzymatic or non-enzymatic pathways in the brain, including neuronal and glial cells, blood-brain ...

15. Transfer Boundary

YOU RECEIVE:

- + Conceptual design + governing equation (Michaelis-Menten + Damköhler)
- + Mechanism description (immobilized NEP on catheter lumen)
- + Bench test protocol (clearance per pass, stability)
- + External precedent catalog (NEP kinetics, immobilization chemistry)
- + This engineering dossier with failure analysis + build plan

YOU MUST DEVELOP:

- > Recombinant NEP production process (GMP-grade)
- > Coating process specification + QC
- > Final catheter geometry (length, diameter, coating density)
- > Manufacturing process for enzyme-active catheter (aseptic assembly)
- > Regulatory submission (likely Class III PMA — no predicate for enzymatic catheter)
- > Clinical validation (CSF A β 42 reduction + clinical benefit endpoint)
- > Long-term stability and reliability data

YOU MUST VERIFY:

- ? Biocompatibility (ISO 10993)
- ? Sterilization compatibility
- ? Mechanical integrity
- ? Regulatory pathway
- ? IP ownership and freedom to operate
- ? Manufacturing process capability

16. Open Questions

- Achievable enzyme surface density Gamma on catheter inner lumen — UNKNOWN
- Enzyme half-life in CSF environment — UNKNOWN
- In vivo clearance per pass — UNKNOWN
- Immunogenicity of immobilized NEP — UNKNOWN
- Substrate competition magnitude — UNKNOWN
- Clinical endpoint for chronic A β 42 reduction — UNKNOWN (no established surrogate)
- Regulatory pathway — UNKNOWN (likely Class III PMA; no predicate for enzymatic catheter)

17. Decisive Next Experiment

WP	Test Article	Equipment	Measurement	Acceptance	Deliverable	Effort
WP-01	Recombinant NEP enzyme	Activity assay (fluorescence substrate)	Activity (k _{cat} /K _M)	Activity > 100% of target	NEP purity > 95%	2-4 weeks
WP-02	NEP-coated catheter	Surface assay (fluorescence)	Surface density (γ)	γ > 10 ¹² units/cm ²	Coating process stable	2-4 weeks
WP-03	NEP-coated catheter	Single flow loop, A β 42 ELISA	Clearance per pass	> 80% reduction per pass	Per catheter design	6 weeks
WP-04	NEP-coated catheter	In vitro chamber, A β 42 assay	Stability (k _{off})	Activity > 50% at target	Stability report	12-24 weeks (chronic)
WP-05	NEP-coated specimen	ISO 10993 test lab	Cytotoxicity, sensitization	ISO 10993 pass	ISO 10993 report	12 weeks (external lab)
WP-06	Coating adhesion specimen	Mechanical test frame	ASTM D3359 adhesion	Adhesion passes ASTM	Adhesion report	6 weeks

18. Buyer Decision Framework

WHY INVEST: Engineering definition exists with 5 governing equations and a 6-step build plan. The next experiment is clearly specified. Target buyer: Biotech company with enzyme therapy program; neuroshunt manufacturer.

WHY WAIT: No physical prototype exists. Regulatory pathway is unresolved. IP ownership is unverified. Manufacturing process is not qualified.

WHY REJECT: Enzyme half-life in CSF environment is less than 30 days (insufficient for chronic implant)

WHAT WOULD CHANGE THE DECISION: A successful bench experiment demonstrating the core mechanism. A qualified regulatory pathway. A confirmed IP position.

19. Kill Condition

KILL_IF: Enzyme half-life in CSF environment is less than 30 days (insufficient for chronic implant)

DISCLOSURE

This dossier distinguishes established evidence, external precedent, computational/modelled results, engineering proposals, and unresolved questions. The absence of physical validation should not be interpreted as evidence that the technology will fail; it means the relevant proposition has not yet been established experimentally.

This dossier does not constitute legal, patentability, FTO, regulatory, or investment advice.